

ARFW0501 · Applied Spatial Analytics

Endterm presentation: Urban Ecology in Delta Cities

Ecological Connectivity and Nature Based Solutions

Rotterdam & Guangzhou

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23-06-2026

Content

Introduction

Analysis

Conclusion/Discussion

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Conclusion/Discussion

Research Focus

Cities : Rotterdam and Guangzhou

Theme : Ecology

RQ: To what extent does green infrastructure connectivity vary across urban areas in delta cities Rotterdam and Guangzhou, and what does this imply for targeted Nature-based Solutions?

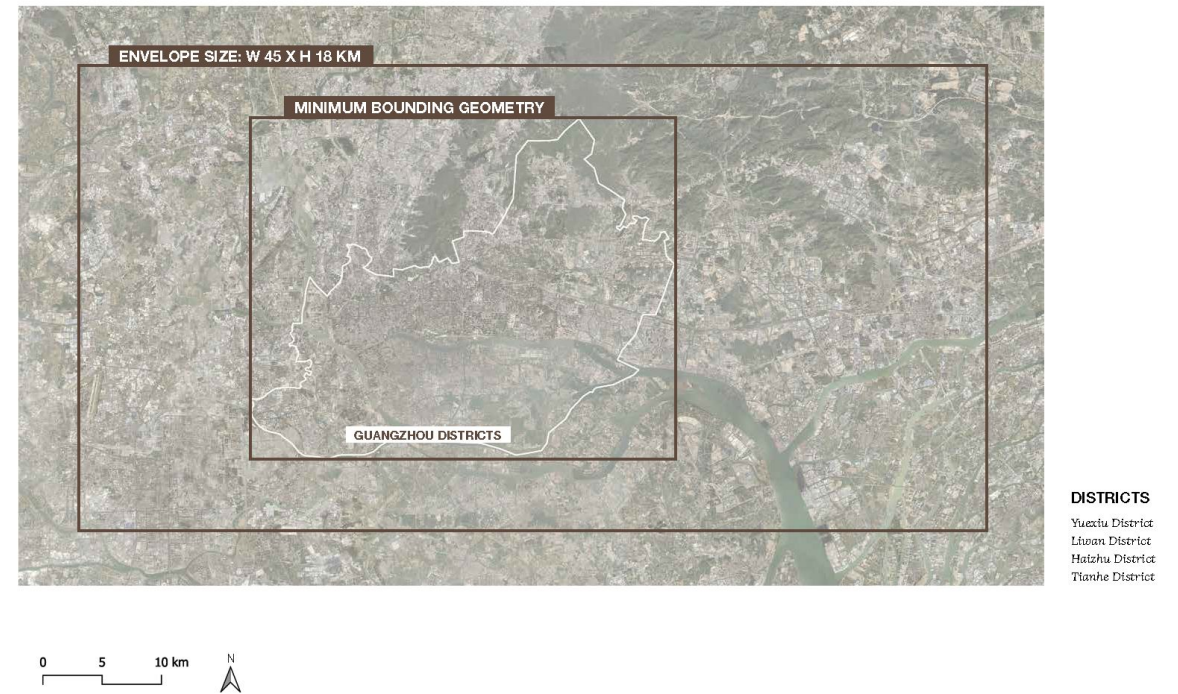
Objective : Explore the current ecological connectivity status in both cities and purpose Nature Based Solutions for ecologically disconnected areas.

Study Areas

Rotterdam



Guangzhou



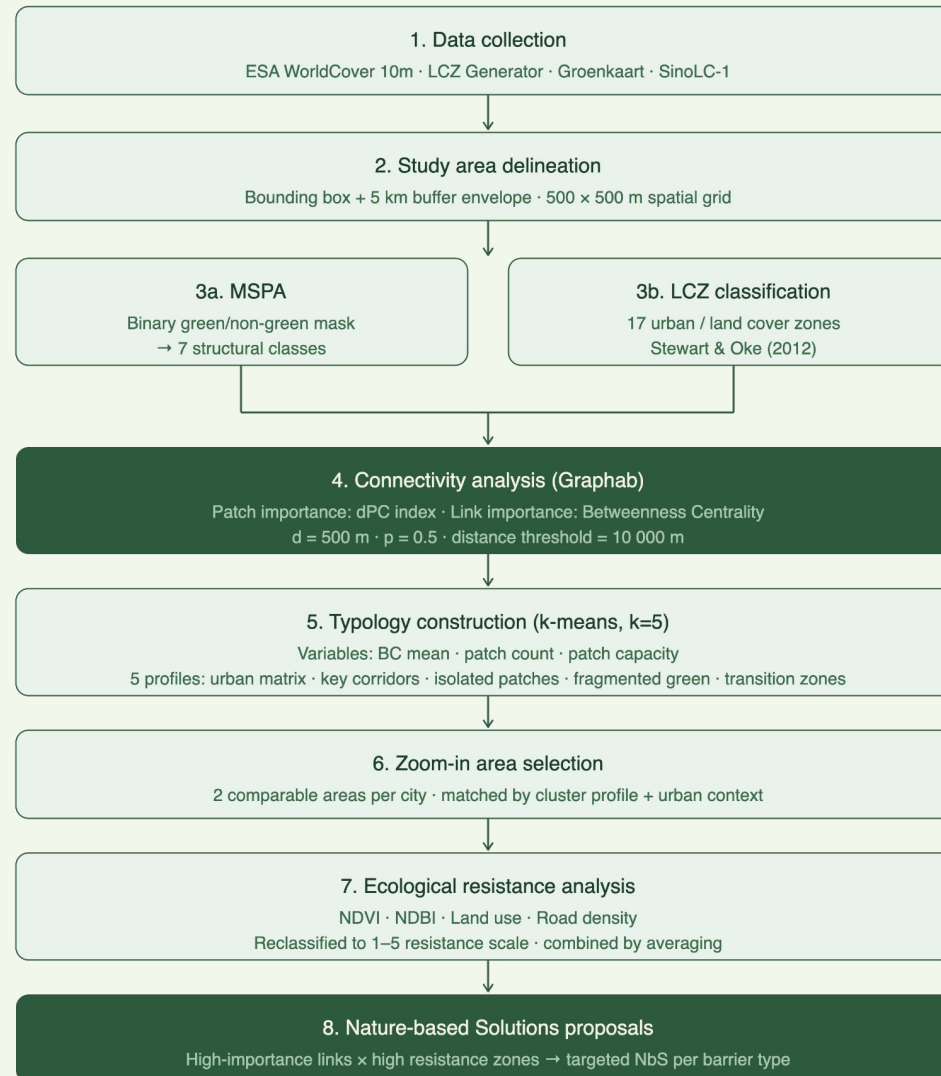
Content

Introduction

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Conclusion/Discussion

Methods Overview



Analysis

MSPA

Patches and Links

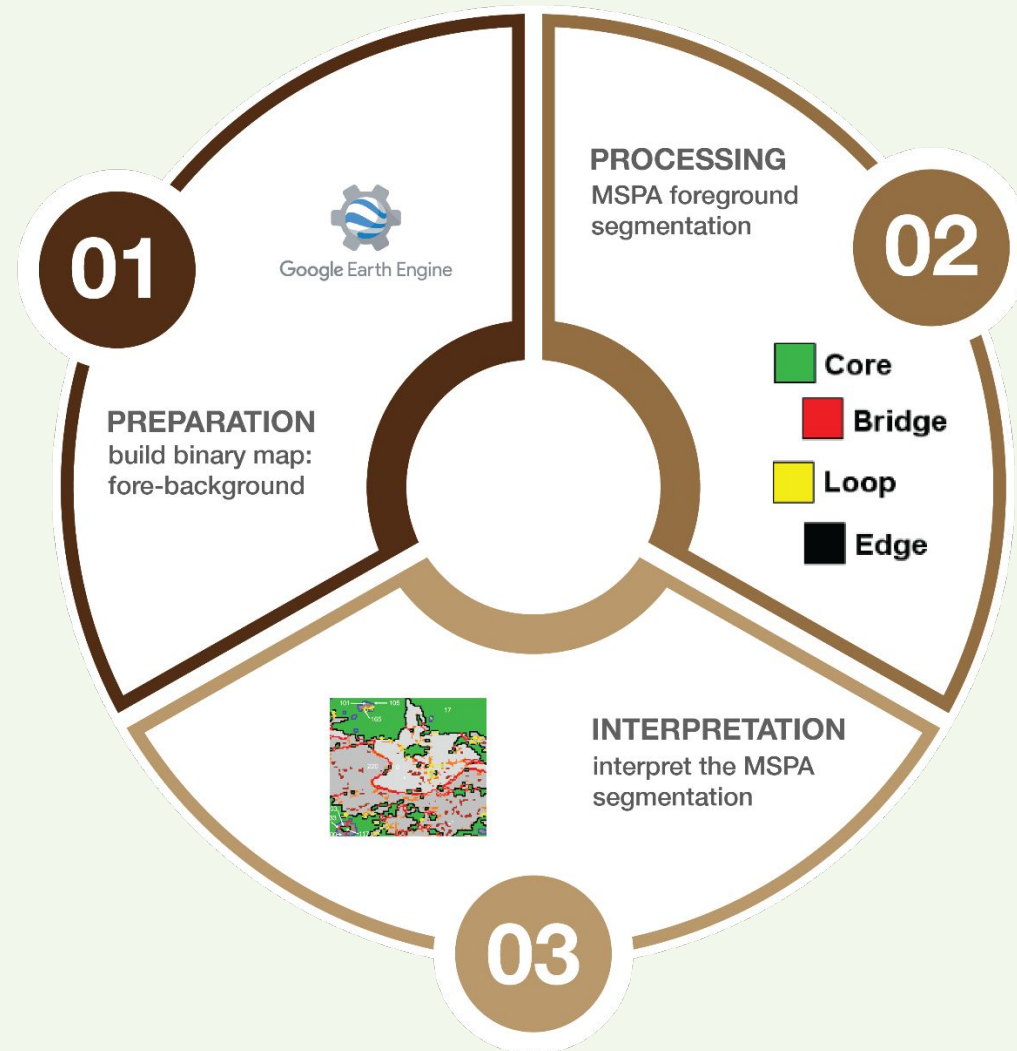
Typologies

Zoom-ins

Ecological Resistance

Morphological Spatial Pattern Analysis

- Understanding the shape and connectivity
- Every pixel assigned to one structural class
- 3 step process



MSPA Rotterdam

ENVELOPE SIZE

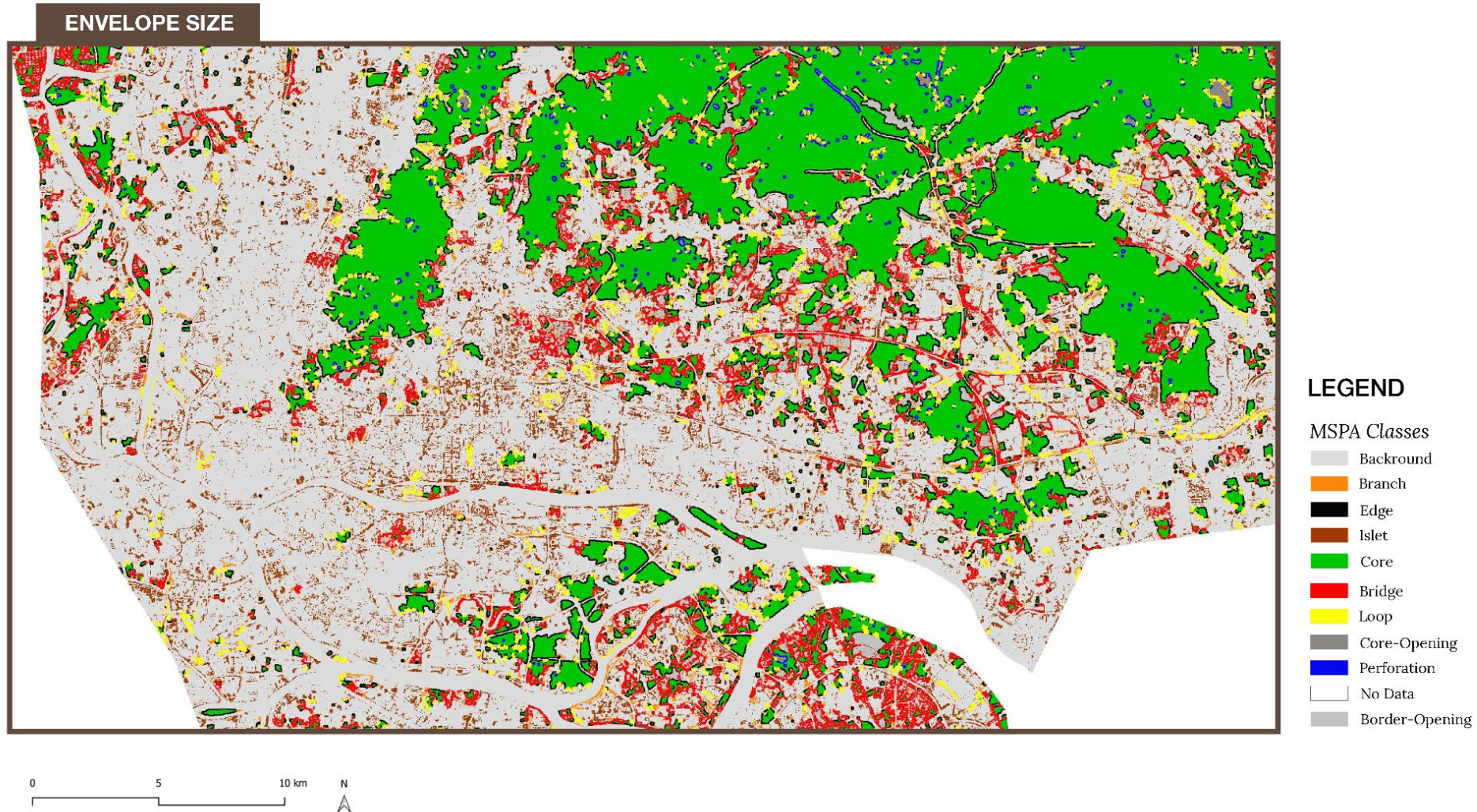


LEGEND

MSPA Classes

- Background
- Branch
- Edge
- Islet
- Core
- Bridge
- Loop
- Core-Opening
- Perforation
- No Data
- Border-Opening

MSPA Guangzhou



Analysis

MSPA

Patches and Links

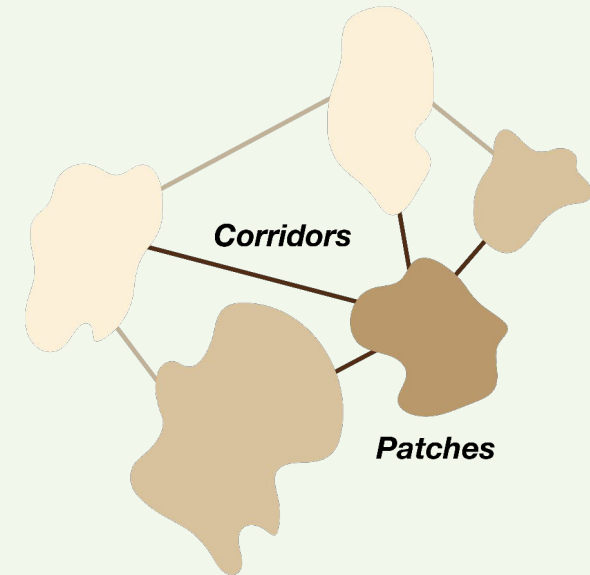
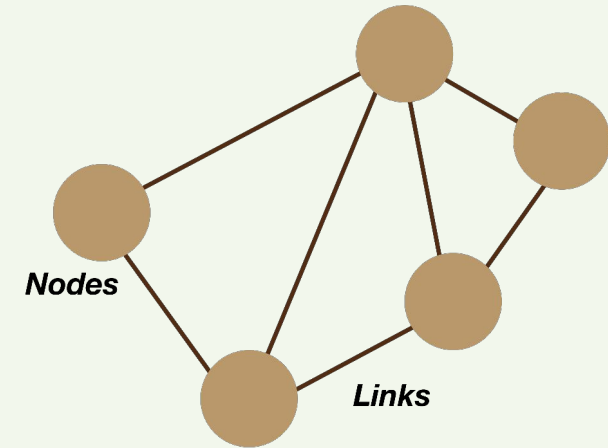
Typologies

Zoom-ins

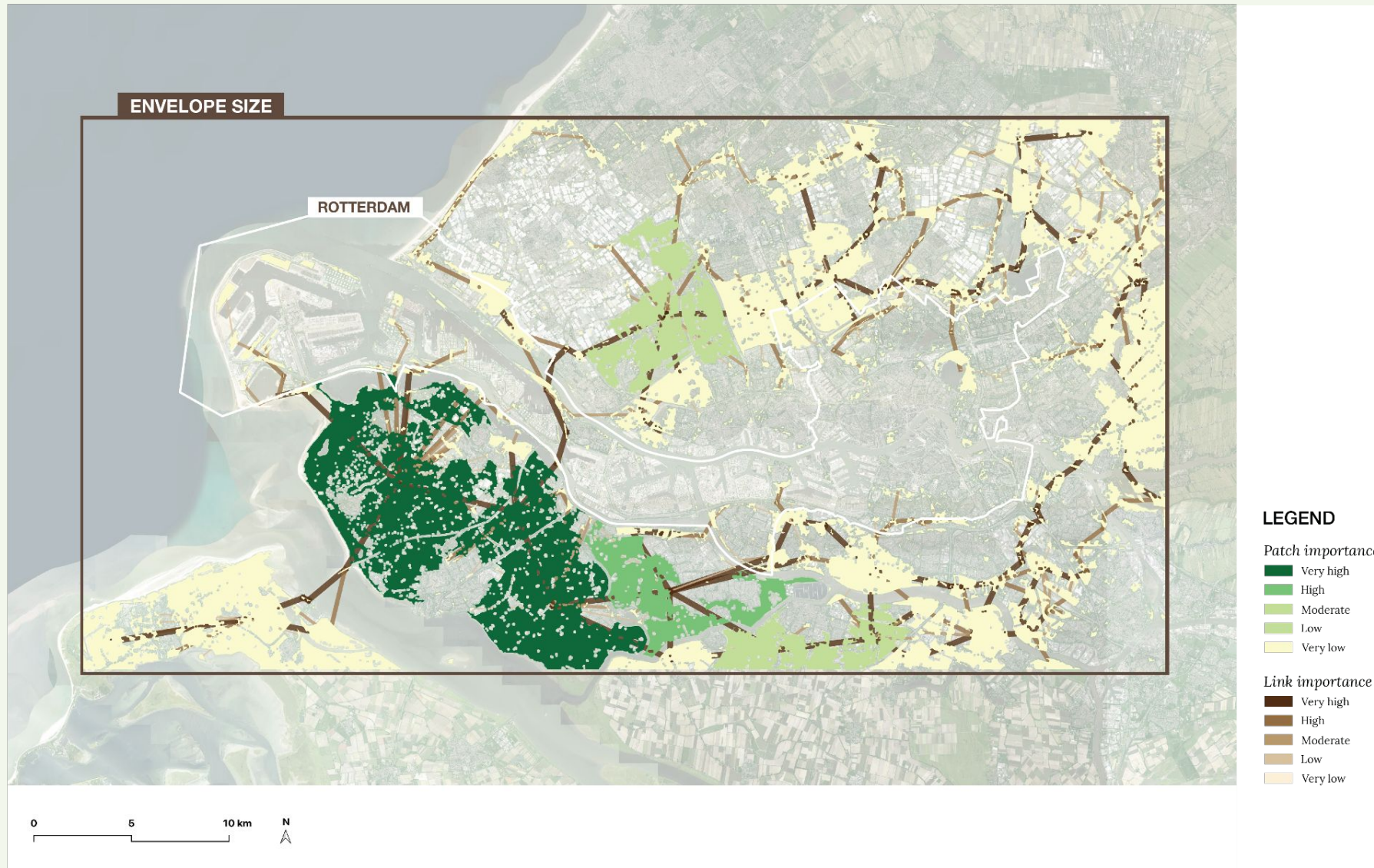
Ecological Resistance

Patch and Link Importance

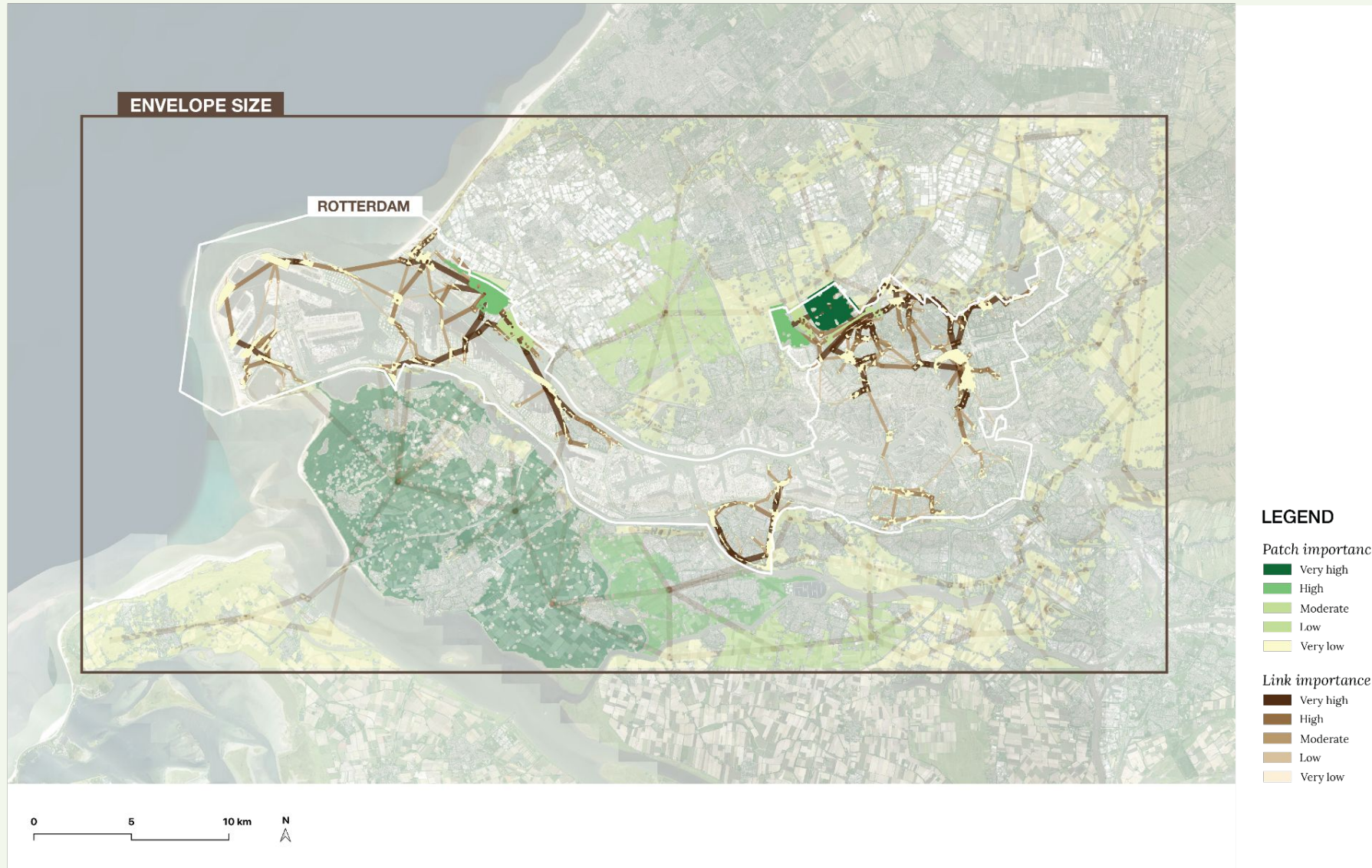
- Graph theory
- delta Probability of Connectivity Index (dPC)
 - Measures patch contribution
 - Connectivity drop for patch removal
- Betweenness Centrality (BC)
 - Counts patch-to-patch routes usage
 - Busiest links
- Dispersal distance (d) and probability (p)
 - $d = 500\text{m}$, $p = 0.5$ (50% chance)



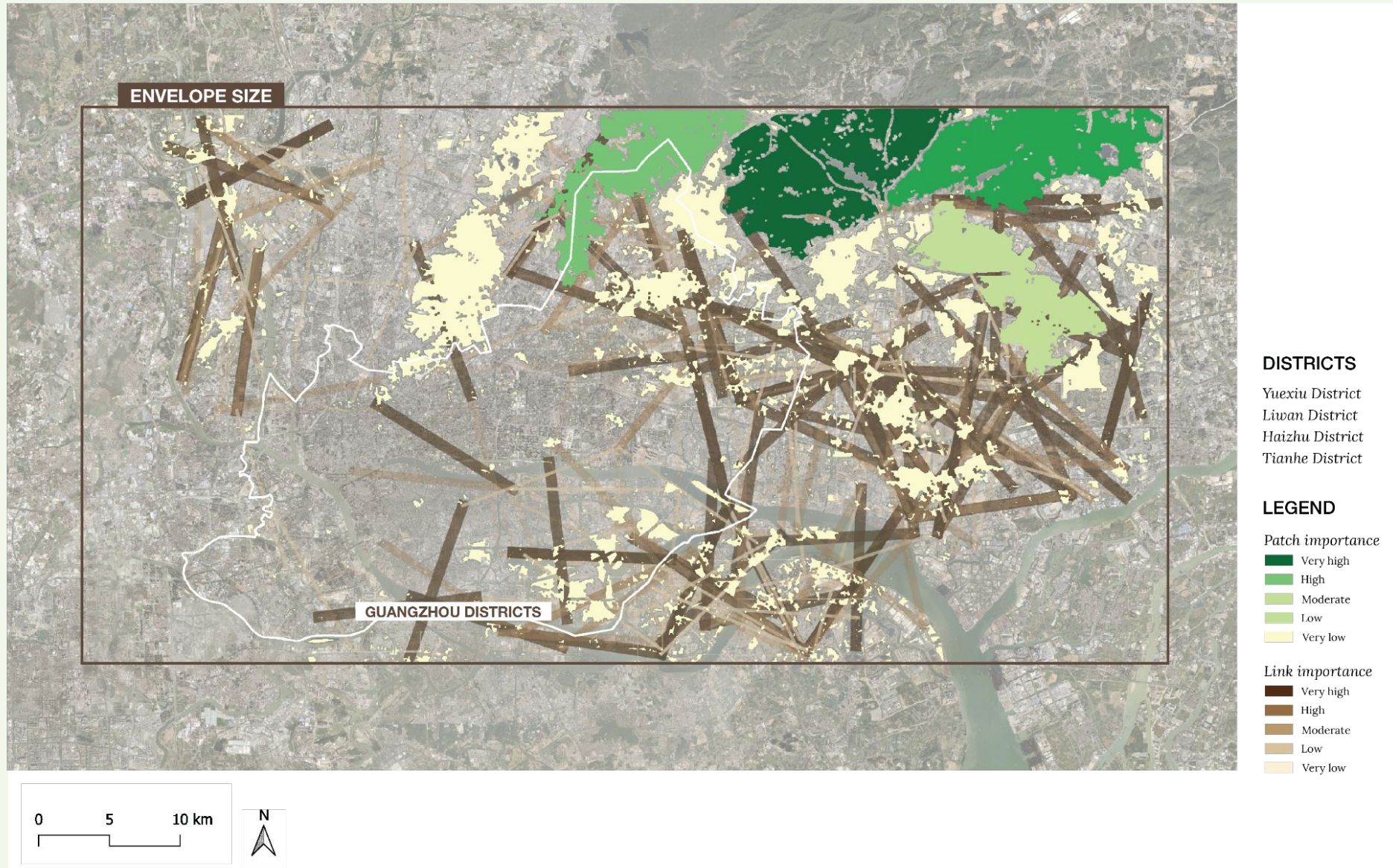
Patch and Link Importance Rotterdam



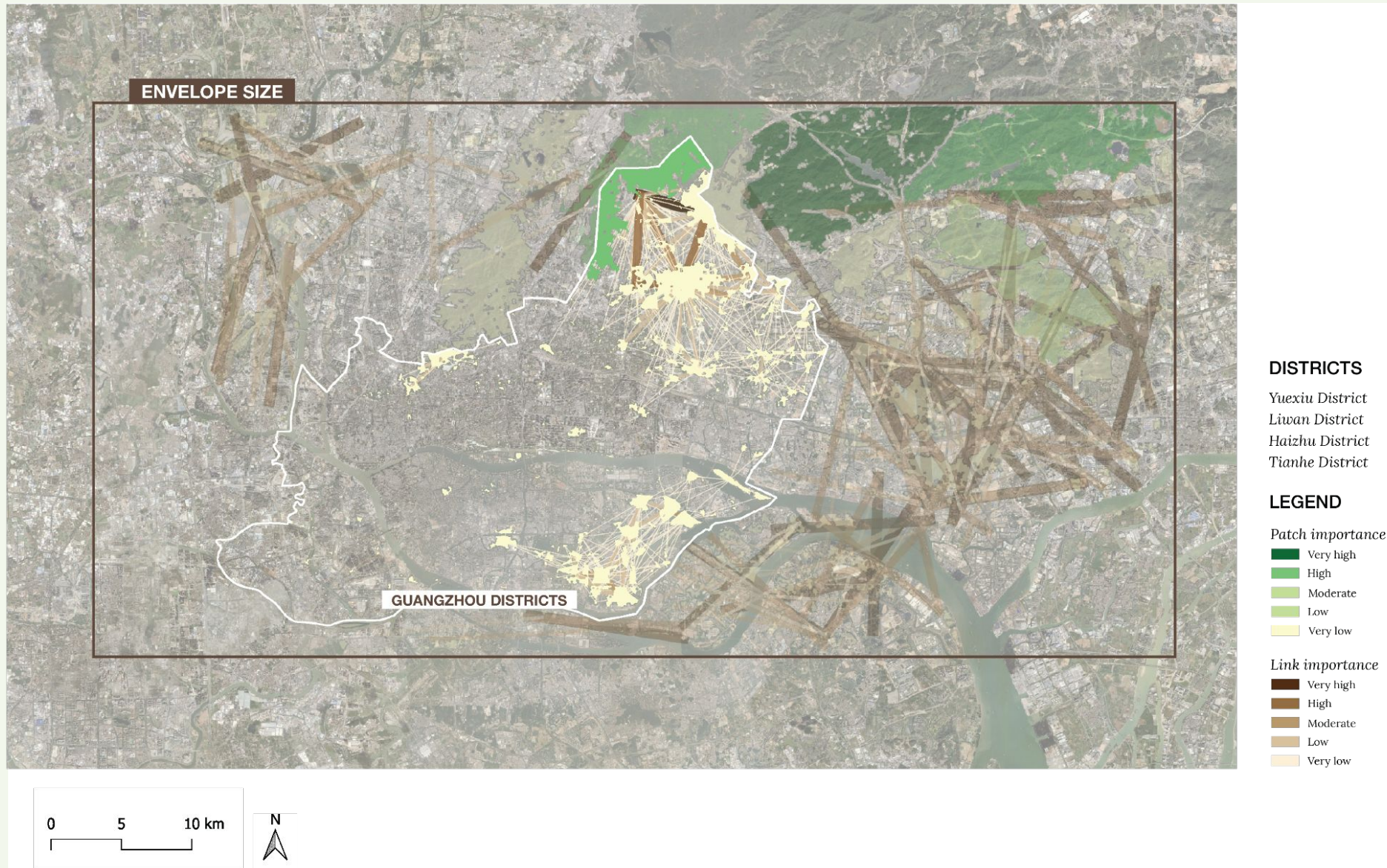
Patch and Link Importance Rotterdam



Patch and Link Importance Guangzhou



Patch and Link Importance Guangzhou



Analysis

MSPA

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Ecological Resistance

Typology Classification: K-means Clustering

- **Variables used:**

Betweenness centrality

Patch count

patch capacity

— measured per 500×500m grid cell

- **K-means (k=5) applied to z-score standardised values**

- **The 5 cluster types:**

 Urban matrix: no green at all

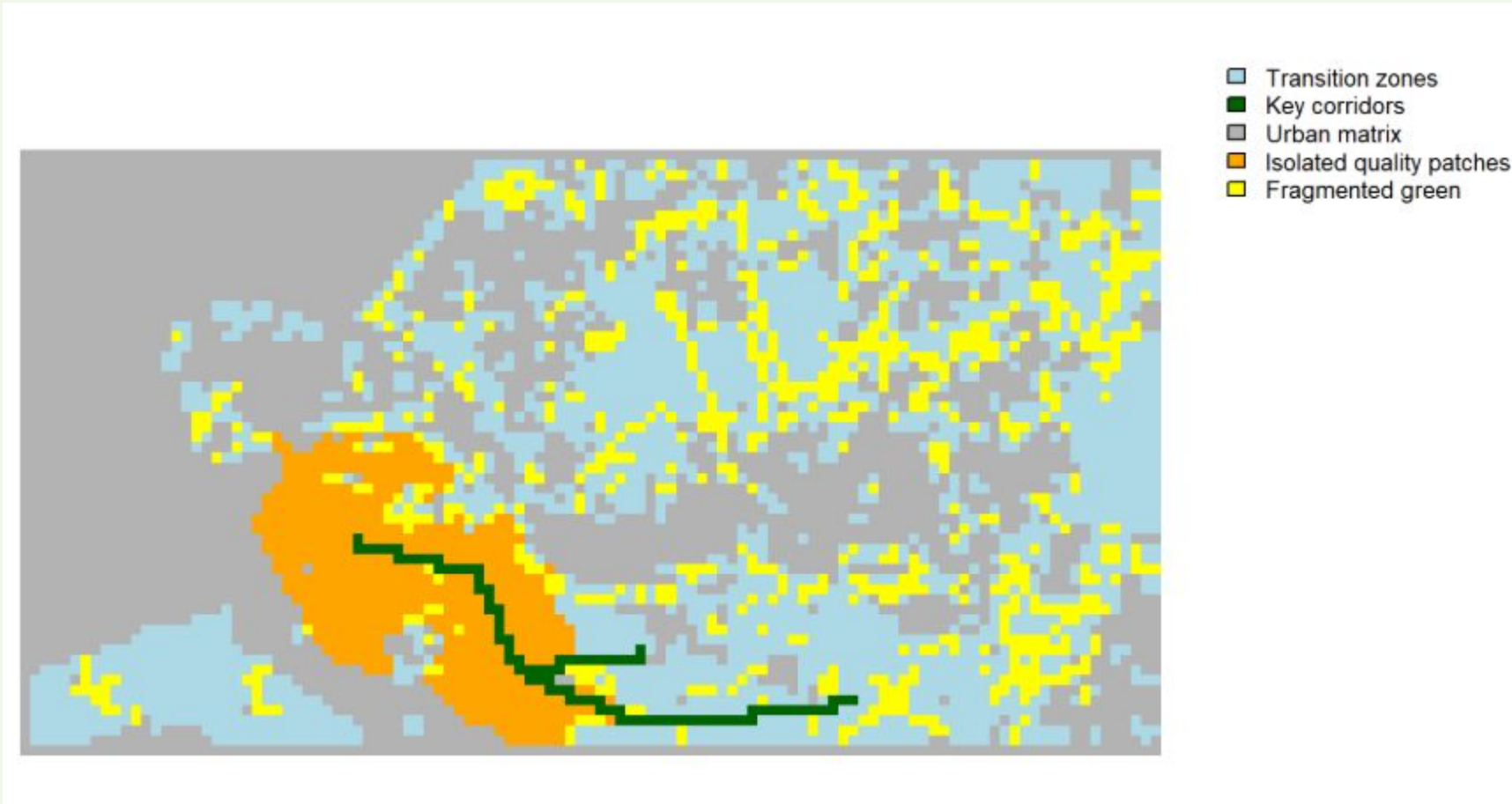
 Key corridors: highest betweenness centrality, good quality patches

 Isolated quality patches: highest patch capacity, poorly connected

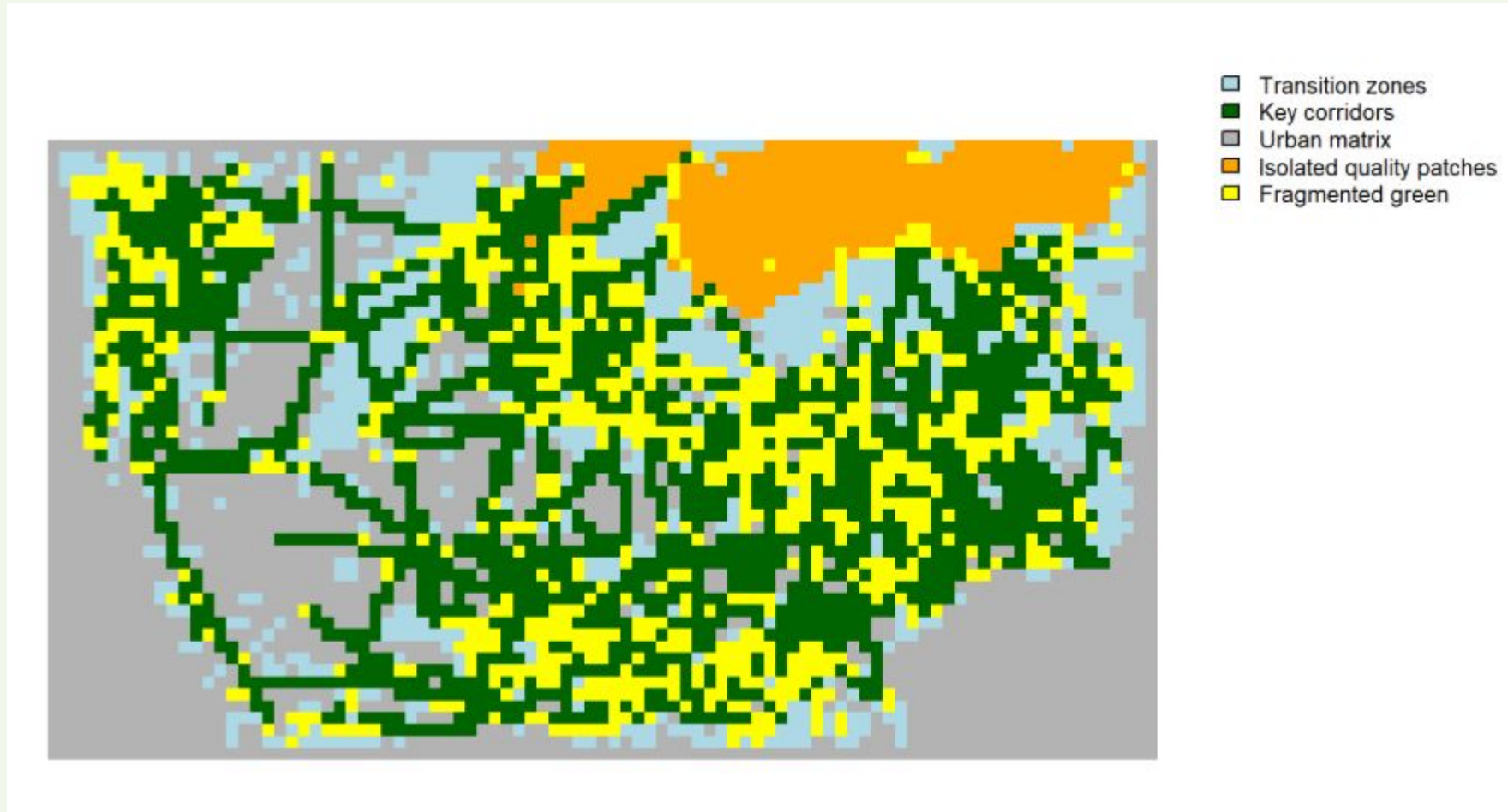
 Fragmented green: many small, low-capacity patches

 Transition zones: weak connectivity, sparse green presence

Typology Classification - Rotterdam



Typology Classification - Guangzhou



Analysis

MSPA

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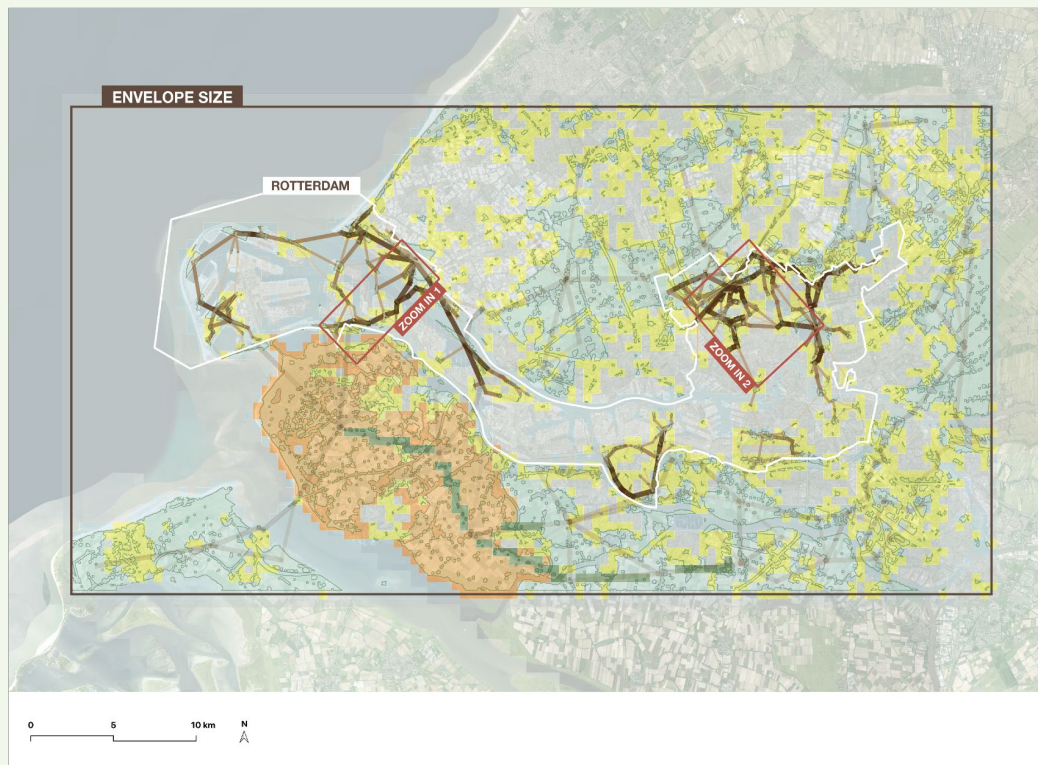
Ecological Resistance

Ecological Resistance

Zoom-in Areas

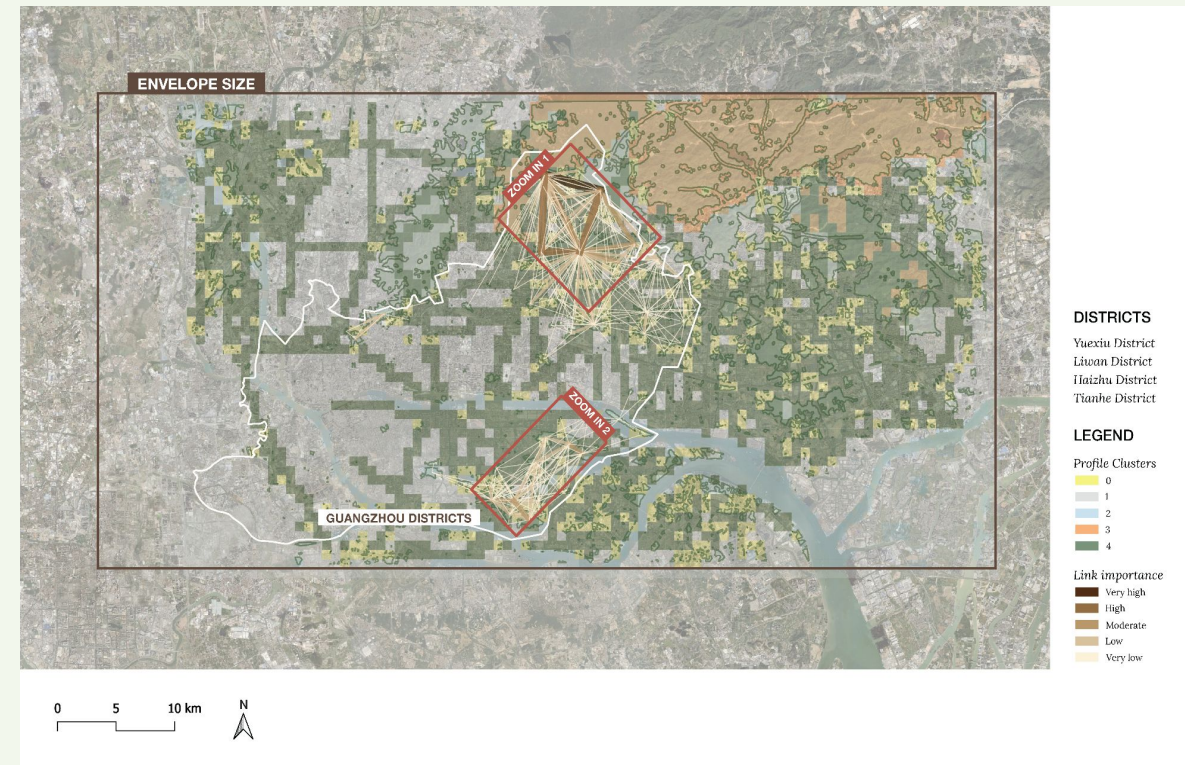
Selection criteria : High-importance links + isolated quality patches + contrasting urban contexts (within municipality/district boundaries)

Rotterdam



- Zoom in 1 : Port area with industrial barriers
- Zoom in 2 : Residential area with transport fragmentation

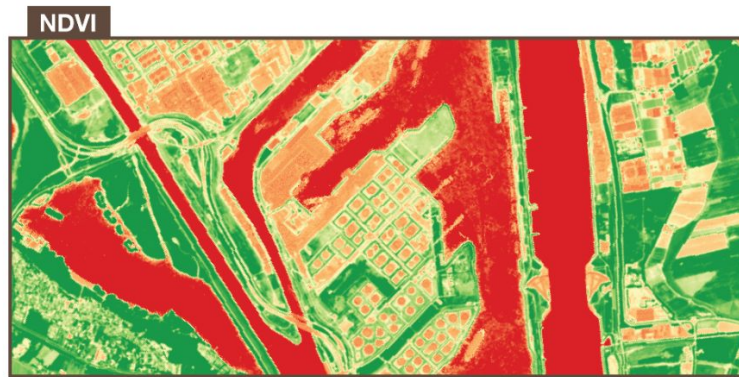
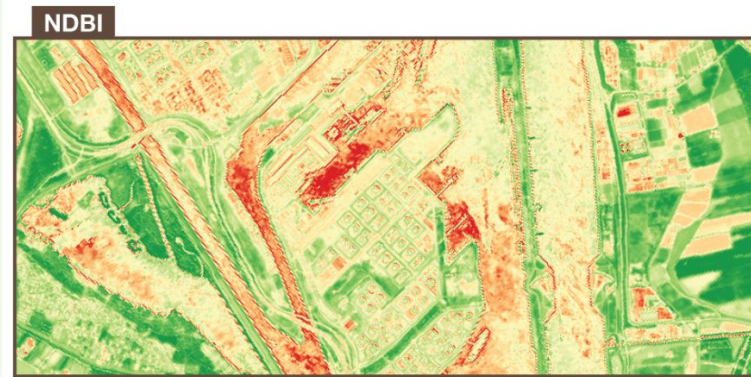
Guangzhou



- Zoom in 1 : Large subtropical forest patches (urban-nature gradient) (Tianhe District)
- Zoom in 2 : Dense urban fabric with river barrier (Haizhu District)

Ecological Resistance

1. $\text{NDVI} + \text{NDBI} + \text{Land cover} + \text{Road density} = \text{Ecological resistance indication}$



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MSPA

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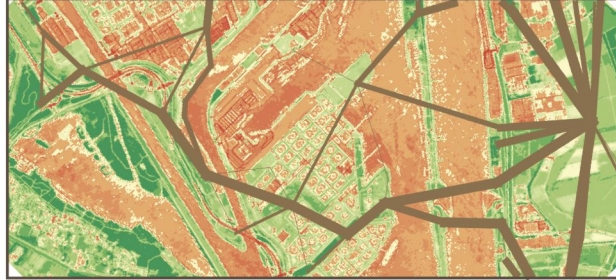
Zoom-ins

Nature-based Solutions

Nature based solutions (NBS)

Zoom-in Rotterdam

ECOLOGICAL RESISTANCE + LINKS



Ecological Resistance

- Very high
- High
- Moderate
- Low
- Very low



ECOLOGICAL RESISTANCE + LINKS



Link importance

- Very high
- High
- Moderate
- Low
- Very low



Barrier and NBS

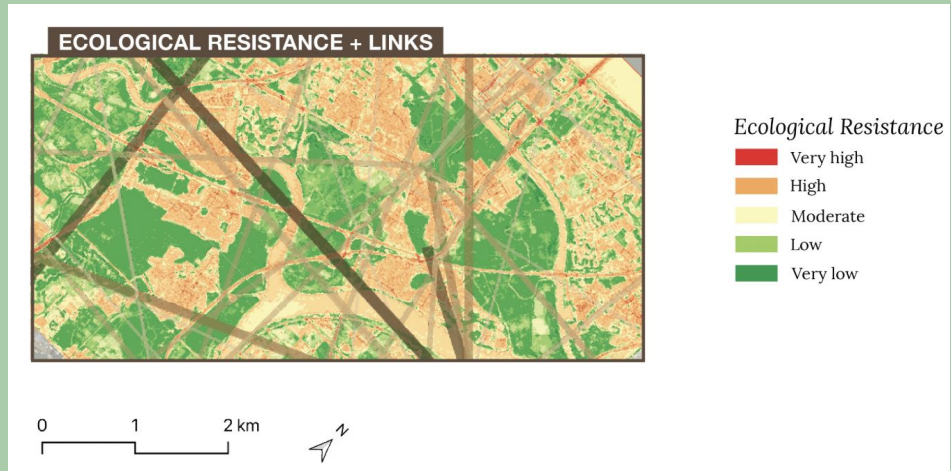
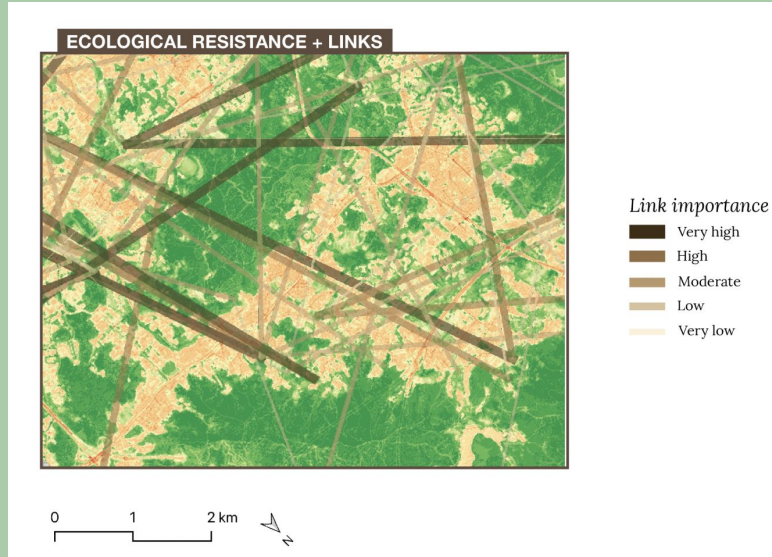
Zoom-in 1 — Port area

Barrier	Proposed NbS
Impervious port surfaces	Green-grey edge retrofitting
Secondary harbour basins	Constructed floating wetlands
Nieuwe Waterweg channel	<i>No NbS feasible — hard barrier</i>

Zoom-in 2 — Residential area

Barrier	Proposed NbS
Roads & railway lines	Green verges & roadside vegetation
Major road/rail crossings	Green bridges & crossing structures
Residential matrix	Pocket parks & street trees

Zoom-in Guangzhou



Barrier and NBS

Zoom-in 1 — Tianhe District

Barrier	Proposed NbS
Roads within urban fabric	Vegetated wildlife underpasses
Urban fabric between forest patches	Green roofs, green walls & pocket parks

Zoom-in 2 — Haizhu District

Barrier	Proposed NbS
River channel & banks	Riparian vegetation restoration
Roads (very high resistance)	Road verge enhancements
Urban fabric between parks	Green roofs & green walls

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Conclusions and discussion

- Rotterdam exhibits a **fragmented, polder-based green network**; Guangzhou a **dense, city-wide corridor system** embedded in urban fabric
- Despite structural differences, both cities share a fundamental connectivity deficit: **ecologically valuable patches remain largely isolated**
- Barrier typology determines intervention logic : waterways, transport infrastructure, and urban fabric each require distinct NbS strategies
- Spatially explicit resistance analysis grounds NbS selection in quantitative evidence rather than qualitative judgment
- Cross-city comparability is methodological in nature; **contextual transferability of findings remains limited**
- Quantitative connectivity analysis can produce spatially targeted intervention priorities independent of species-level data